

Automatic Detection of Scoria Cones from DEMs

Gabriele Berruti (4791919)
Computational Vision — University of Genoa

1 Introduction

This project detects *scoria cones* from single-channel Digital Elevation Models (DEMs), where each pixel encodes a terrain elevation in metres. The core contribution is the **separation of nested or adjacent cones**: point-based blob detectors collapse two overlapping craters into a single response, and a one-to-one matching then discards one of the two. We address this with a watershed segmentation of the crater depressions, which natively splits adjacent catchment basins.

We work on three datasets: **El Hierro** (5 m resolution, with ground-truth annotations, used for development and validation), **Etna** (2 m, noisier ground truth, used as a stress test) and **Jeju** (10 m, no ground truth, used for qualitative and unsupervised analysis).

2 Data and preprocessing

2.1 Step 0 — DEM loading and inspection

A DEM is a *range image*: a single elevation band. Each dataset uses a different *nodata* sentinel, which we convert to `NaN` so that all later computations ignore invalid pixels naturally. Table 1 summarises the three DEMs after loading.

In Table 1, $H \times W$ is the raster size in pixels (matrix rows $\times$ columns), Mpx the total number of pixels in millions ($H \cdot W$, a proxy for the memory footprint), Res. the ground size of one pixel in metres, dtype the numeric type of the elevation values, and $z_{\max}$ the maximum elevation (sanity check against the real volcano).

Dataset	Size (H×W)	Mpx	Res. (m)	dtype	$z_{\max}$ (m)
El Hierro	5081 × 5681	28.9	5	float32	1499.8
Etna	27756 × 25717	713.8	2	float32	3324.1
Jeju	4364 × 7761	33.9	10	float32	1945.0

Table 1: DEM inspection. Maximum elevations match the real volcanoes (Etna ≈ 3.3 km, El Hierro ≈ 1.5 km, Hallasan/Jeju ≈ 1.95 km), confirming correct loading.

Three facts drive the following steps: (i) Etna is ≈ 714 Mpx (≈ 2.86 GB as float32), so it must be processed in *tiles* rather than loaded whole; (ii) the three *nodata* conventions differ (-3.4×10^{38} , -9999 , `NaN`) and are all mapped to `NaN`; (iii) each dataset has its own UTM coordinate reference system, so any shapefile must be reprojected into the CRS of its DEM before use.

2.2 Step 0.5 — Visual inspection

Before deriving any channel we inspect the data on El Hierro. Figure 1 shows that the raw DEM looks visually flat, whereas the Eduard renderings (NE/NW shading and texture shading) reveal the

cones; this motivates the derived channels of Step 1. Figure 2 shows the 196 crater outlines of the ground truth: the two closest craters lie only 25 m apart (5 pixels) and several cones are adjacent or nested. Point-based blob detectors merge such pairs into a single detection — separating them is the goal of the watershed step.

For clarity, the panels in Figure 1 are: the **raw DEM**, i.e. the original elevation grid where each pixel is a terrain height in metres (it looks flat because a cone is only tens of metres tall relative to the whole island); the **Eduard renderings**, relief-shaded views produced by the *Eduard* relief-shading tool and shipped with the dataset; the **NE/NW shading** (hillshade), where the terrain is lit by a simulated sun from the North-East/North-West so that slopes facing the light are bright and those facing away are dark (two opposite directions are given so no slope stays in shadow); and the **texture shading**, a light-direction-independent rendering that emphasises fine surface detail.

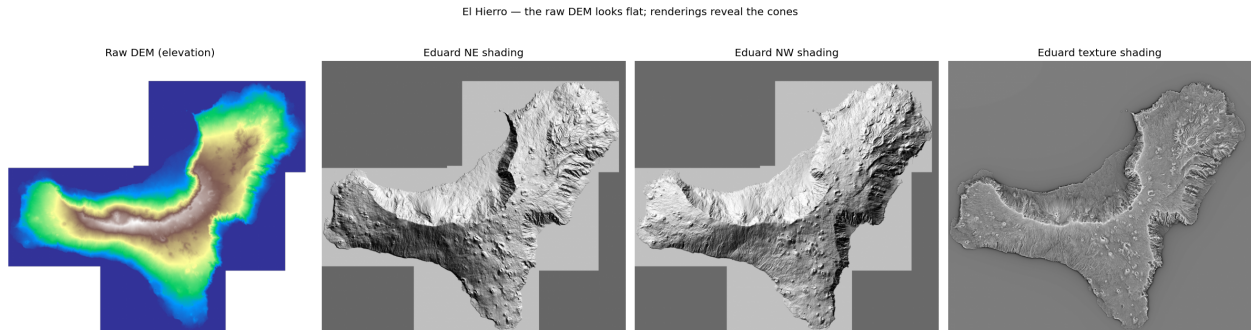


Figure 1: El Hierro: the raw DEM (left) is visually flat, while the Eduard renderings reveal the conical landforms.

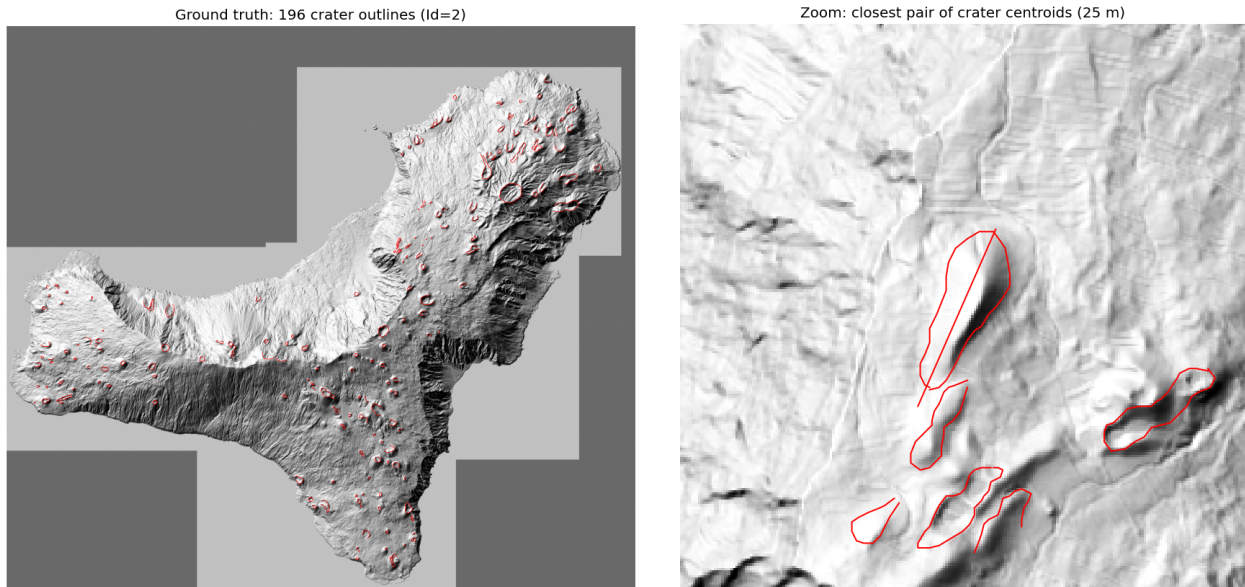


Figure 2: Ground truth on El Hierro. Left: the 196 crater outlines (Id= 2). Right: a zoom on the closest pair of crater centroids (25 m apart).

Figure 3 isolates the cones that actually *overlap*, i.e. whose nearest neighbour is closer than the sum of the two radii. These are 8 craters in 4 pairs; note that they are not the single closest

centroids of Figure 2 (two small craters), but pairs of *large* adjacent cones whose outlines intersect. These are the configurations a point-based detector merges into one and the watershed step must separate.

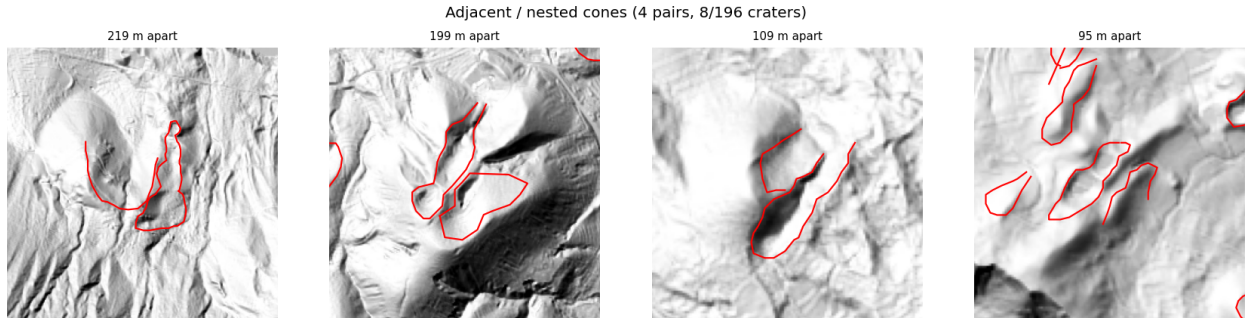


Figure 3: The adjacent/nested cases on El Hierro: the 4 crater pairs whose outlines overlap (8 craters). The watershed step is designed to split exactly these.

Ground-truth quality and label semantics. The El Hierro shapefile is *dirty* and mixes three feature types in a single file. The crater outlines (Id= 2) are open polylines: 191 of 196 do not close (median start–end gap 40 m, up to 556 m) and some have as few as 2 vertices. We therefore use crater *centroids* (plus an approximate radius) rather than the exact outline shape. A geometry test then characterises the three labels (Table 2): Id= 1 is the cone **base** (large, round, and enclosing the crater in 98% of cones), Id= 2 the **crater** rim, and Id= 3 a **fissure** — a short, strongly elongated segment that always carries a non-zero azimuth in the `Prj_Az` field. The `pericolosi` field is a hazard level (0/1/2). We adopt Id= 2 as the detection target.

Id	meaning	n	verts	length (m)	aspect	Prj_Az≠0
1	base	151	29	846	1.29	0%
2	crater	196	19	457	1.47	0%
3	fissure	171	2	272	8.32	100%

Table 2: Data-driven characterisation of the El Hierro labels (median values). The high aspect ratio and the always-present azimuth identify Id= 3 as fissures.

2.3 Step 0.6 — Quick data analysis

Three cheap measurements on El Hierro parameterise the later steps rather than guessing their constants. (i) The three Eduard renderings share the DEM grid exactly (same 5081×5681 shape and transform), so overlays are pixel-exact. (ii) The equivalent radius of the crater outlines (Figure 4, left) has median 71 m and a 5th–95th percentile range of 20–174 m, which we adopt as the LoG detector scale range. (iii) The nearest-neighbour distance between craters (Figure 4, right) has median 418 m but a minimum of 25 m, and 8/196 craters (4%) lie closer to a neighbour than the sum of the two radii, i.e. they are adjacent or nested. This 4% is precisely the population that point-based detection merges and that the watershed step is designed to recover.

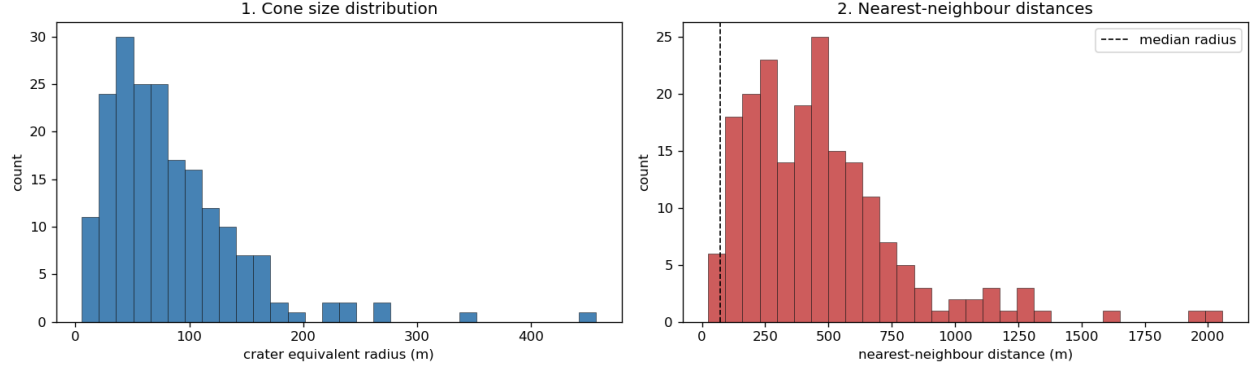


Figure 4: El Hierro crater statistics. Left: equivalent-radius distribution, used to set the LoG scale range. Right: nearest-neighbour distance distribution; the dashed line is the median radius, and the cones below it are the adjacent/nested cases.

3 Detection pipeline

3.1 Step 1 — Morphometric channels

From the single elevation band we derive three channels with the image gradient (deck 01_image_processing, p.13): the **slope** = $\arctan \sqrt{g_x^2 + g_y^2}$ in degrees, the **aspect** = $\text{atan2}(g_y, g_x)$ (the direction the slope faces), and the **roughness**, the local standard deviation of elevation in a 5×5 window. Figure ?? shows them on El Hierro. As a quantitative check, the median slope on the crater rims is 18.8° against 8.1° on the background (about $2.3\times$ steeper), and the aspect rotates around each cone, giving the radial pattern of the rightmost panel. These three channels, together with the texture rendering, are the input of the detection steps.

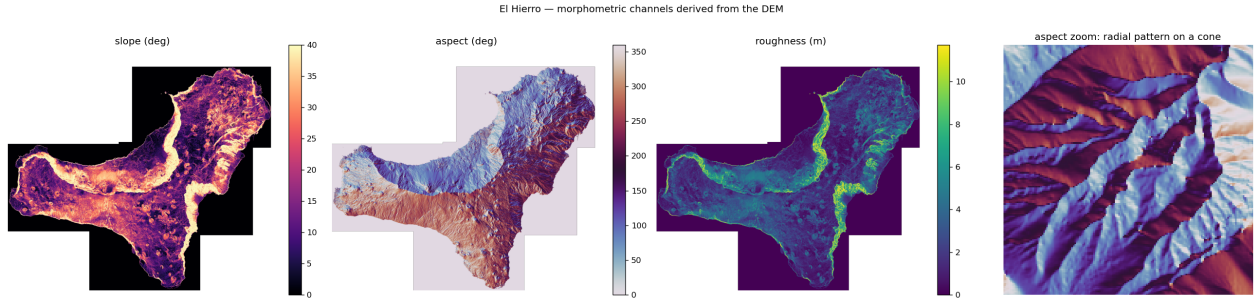


Figure 5: Morphometric channels derived from the El Hierro DEM: slope, aspect and roughness, plus a zoom of the aspect showing the radial pattern around a single cone.